

LYSIUK VLADYSLAV PAVLOVYCH

Electronics design engineer (Hardware / PCB Design)



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Date of birth: 01.04.2005

About me

Junior electronics design engineer. Experience across the full device development cycle: design of electrical schematic diagrams, creation of component libraries in Altium Designer, placement and routing of printed circuit boards, assembly and soldering of prototypes, bring-up, test firmware for STM32 and Arduino Uno microcontrollers. Simulation in Micro-Cap, Ansys (PyAEDT) and MATLAB. Participant of the international projects Erasmus+ (DIGITRANS), DAAD Eastern Partnerships and Horizon Europe (NGI Search). Bachelor's degree with honours.

Education

Bachelor (degree with honours) — 172 Telecommunications and Radio Engineering

09.2022 – 06.2026

Chernihiv Polytechnic National University, Chernihiv, Ukraine

- Topic of the final qualification work: "Solenoid valve control system with wireless data exchange".
- Within the diploma I performed circuit calculations, printed circuit board routing, microcontroller firmware and simulation in MATLAB (the circuit design came from the academic supervisor, the work was carried out under his guidance).
- Level 6 EQF / NQF.

Project, research and engineering experience

Hardware part of the remote power electronics laboratory (LabDiscoveryEngine)

2023 – 2024

Chernihiv Polytechnic National University (within the NGI Search project, Horizon Europe, project No. 101069364)

- Creation of component libraries and drawing of electrical schematic diagrams in Altium Designer.
- Placement and routing of the power electronics printed circuit board under a lecturer's supervision.
- Assembly and soldering of the boards; the assembled benches are still in use in the university's teaching process.

The results of the work were published in the proceedings of the conference "Youth Science — 2024", page 1039 <http://ir.stu.cn.ua/handle/123456789/30262>.

Hardware part of the remote microprocessor technology laboratory (Erasmus+ DIGITRANS)

2024 – 2025

EACEA, Erasmus+ Capacity Building, project No. 101127683 — DIGITRANS

- Development of a shield (expansion board) for the STM32 NUCLEO-H723 bench: component libraries, work with schematics, independent placement and routing of the board.
- Testing and soldering of the boards.
- Assembly and connection of the benches (Nucleo + Raspberry Pi + Analog Discovery); the benches are in use at the university.
- Test firmware for the microcontroller: LED control and output of information to a 7-segment display with

multiplexing driven by external signals.

- The results were published in the proceedings of the conference NTSS-2024, page 102
<http://ir.stu.cn.ua/handle/123456789/32382>

Performer profile: <https://hub.stu.cn.ua/project-performer/lysyuk-vladyslav-pavlovych/>

Multichannel thermocouple temperature measurement system (DAAD) 24.10.2024 – 24.12.2024
DAAD Eastern Partnerships 2023–2025, “Student Exchange with Online Participation” scholarship

- Simulation of the analogue path (switching, amplification of the signal of J/K type thermocouples) in Micro-Cap.
- Calculation of the switching, amplification and filtering circuits, selection of components, verification of filtering and switching time under laboratory conditions.
- Breadboarding of a 2-channel prototype on a breadboard, control and testing via Arduino.
- Reworking of the schematic and the libraries of the existing project of the 24-channel measurement board; the following year — independent routing of the board in Altium Designer.
- The results were published in the proceedings of the conference “Youth Science — 2025”. Page 1012
<http://ir.stu.cn.ua/handle/123456789/32545>

Performer profile: <https://hub.stu.cn.ua/project-performer/lysyuk-vladyslav-pavlovych/>

International internship — Hochschule Bonn-Rhein-Sieg (H-BRS) 11.2025 – 12.2025
University of Applied Sciences Bonn-Rhein-Sieg, Sankt Augustin, Germany (within DAAD Eastern Partnerships, project No. 57634194)

- International internship programme (07.11.2025 – 07.12.2025), certificate of successful completion.
<https://drive.google.com/file/d/1uHTCfVhg4tousyQB5vbU6JxceFJbixON/view?usp=sharing>
- Routing of the multichannel temperature measurement board in Altium Designer, routing of the board for the qualification work and related engineering tasks.

Pre-diploma internship — Hochschule Bonn-Rhein-Sieg (H-BRS) 05.2026 – 06.2026
University of Applied Sciences Bonn-Rhein-Sieg, Sankt Augustin, Germany (DAAD “Eastern Partnership”, project No. 57634194)

- Pre-diploma internship within short-term academic mobility (02.05 – 10.06.2026), DAAD grant.

Research and development in the field of power electronics 2025 – 2026
Department of Electronics, Automation, Robotics and Mechatronics

- Research on a control algorithm for solenoid valves (three-phase actuation profile) based on STM32H743; the results were published in the proceedings of the conference “Youth Science — 2026”. <https://ir.stu.cn.ua/handle/123456789/35067> page 1094
- Routing of the board of the SIC transistor gate driver module and simulation of the parasitic capacitance of the printed circuit board traces in Ansys. The results were published at KZYATPS-25, volume 3, page 19
<https://drive.google.com/file/d/1juZTLLvSuyn-di0HTmq1ew8Ju6fRK2ki/view>.
- Co-author of a study of ferromagnetic shielding in wireless power transfer systems (simulation via PyAEDT/Ansys), publication at the conference “Comprehensive quality assurance of technological processes and systems”, 2025.
- Performer of the project “Simulation and optimisation of activation capacitors for hybrid dynamic wireless power transfer systems” (competition of research works of young scientists).
- Bachelor’s qualification work: “Solenoid valve control system with wireless data exchange” — independent development of the board, the controller firmware and the MATLAB model (see the section “Education”).

Publications

- V. P. Lysiuk and M. A. Khomenko, “Multichannel temperature measurement using J and K type thermocouples,” in XV International Scientific-Practical Conference of students, postgraduates and young scientists “YOUTH SCIENCE — 2025”, Chernihiv, Ukraine: Chernihiv Polytechnic National University, 2025, p. 1012.
- V. P. Lysiuk, M. A. Khomenko, “Hardware part of the remote power electronics laboratory based on the LabDiscoveryEngine system” — Youth Science — 2024, Chernihiv, 2024, p. 1039.

- V. P. Lysiuk and M. A. Khomenko, "Hardware part of the remote laboratory of microprocessor technology," in The latest technologies of modern society NTSS-2024, Chernihiv, Ukraine: Chernihiv Polytechnic National University, 2024, p. 102.
- B. Pakhaliuk, V. Lysiuk, and O. Husev, "Analysis of ferromagnetic shielding patterns in wireless power transfer applications using the PyAEDT library in Ansys," in Comprehensive quality assurance of technological processes and systems, May 2025.
https://www.researchgate.net/publication/392126449_Analysis_of_ferromagnetic_shielding_patterns_in_wireless_power_transfer_applications_using_the_PyAEDT_library_in_Ansys
- B. Pakhaliuk, V. Lysiuk, I. Burmaka, O. Matiushkin, and R. Strzelecki, "Simulation analysis of wireless power transfer solutions with hybrid inductive and capacitive coupling," in XX International Conference MODS 2025: Mathematical and Simulation Modelling of Systems, 2025.
- L. A. Yarmolenko, V. P. Lysiuk, and B. P. Pakhaliuk, "Experimental determination of the growth rate of a copper electrolytic coating for a 3D printed air capacitor," in The latest technologies of modern society (NTSS-2025), Chernihiv, Ukraine, 2025.
- V. Lysiuk, B. Pakhaliuk, A. Strohii, L. Yarmolenko, "Development of an optimized PCB topology for a SiC gate driver in a hybrid dynamic wireless power transfer system," Chernihiv Polytechnic National University, Chernihiv, 2026, p. 19.
- V. P. Lysiuk, M. A. Khomenko, "Multichannel solenoid valve control device" — Youth Science — 2026, Chernihiv, 2026.

Grants and international programmes

- Horizon Europe, NGI Search (HORIZON-CL4-2021-HUMAN-01), project No. 101069364 — participant in the development of the hardware part.
- Erasmus+ (EACEA), project No. 101127683 DIGITRANS — development of the equipment of the remote laboratory.
- DAAD Eastern Partnerships 24.10.2024 – 24.12.2024 — "Student Exchange with Online Participation" scholarship (900 EUR).
- DAAD "Eastern Partnership", project No. 57634194 — academic mobility to Hochschule Bonn-Rhein-Sieg (07.11.2025 – 07.12.2025, grant 861 EUR).
- DAAD "Eastern Partnership", project No. 57634194 — academic mobility to Hochschule Bonn-Rhein-Sieg (02.05 – 10.06.2026, grant 1342 EUR).

Awards

- Bachelor's degree with honours (2026).
- Letter of appreciation from the Mayor of Chernihiv (May 2024) — for a significant personal contribution to strengthening cooperation at the European level and to the development of democratic values, on the occasion of Europe Day (<https://rtes.stu.cn.ua/2024/05/16/europa-day-event/>).

Technical skills

- Circuit design and layout: development of electrical schematic diagrams; Altium Designer (component libraries, schematics, placement, board routing, including power electronics).
- Simulation: Micro-Cap, Ansys / PyAEDT (electromagnetic simulation, parasitic parameters), MATLAB/Simulink.
- Embedded systems: STM32, Arduino; C for microcontrollers; work with UART, PWM, ADC, I2C, SPI.
- Hands-on: assembly and soldering (THT/SMD), breadboarding, with the use of measurement equipment.

Languages

	Understanding		Speaking		Writing
	Listening	Reading	Spoken interaction	Spoken production	
Ukrainian	C2	C2	C2	C2	C2
Russian	C2	C2	C2	C2	C2
English	B2	B2	B1	B1	B1